



Sandhya Rai

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Computer Science and Engineering

B.Tech

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EDUCATION

•National Institute Of Technology, Patna	2022-2026
B.Tech in Computer Science and Engineering	6.79/CGPA
•RPS School, Bihar	2020-2022
central Board of secondary Education , Bihar	68.2 Percentage
•ST.Paul's English school Harnaut	2018
central Board of secondary Education , Bihar	66.5 Percentage

EXPERIENCE

•Prodigy Infotech	16/08/2024
Full-Stack Web Development	Online
– I have worked on full-stack web development and learned how to apply skills on project.	

PERSONAL PROJECTS

–Dev Hospital Management System

(Improves accessibility, patient management, emergency response, appointment scheduling, security, and doctor verification)

- * Tools technologies used: HTML, CSS, JavaScript, ReactJS.
- * – Easy Access – Patients can book appointments online access emergency services instantly. Efficient Management – Doctors staff can add/update patient records easily. Faster Emergency Response – Emergency button allocates doctors quickly. Simplified Appointments – Patients can see doctor availability schedule visits hassle-free. Secure Authentication – Login/Register system ensures authorized access. Resume Upload for Doctors – Helps hospitals verify doctor qualifications..)

User Allocation in Edge Computing Using Quantum-Inspired Cat Swarm Optimization (QCSO) with a Novel Hashing Technique

6th Semester / National Institute of Technology Patna

Technologies: Python, Quantum-Inspired Differential Evolution, Metaheuristics

Description:

- Proposed a hybrid optimization algorithm combining Cat Swarm Optimization (CSO) with quantum principles for efficient user allocation in Edge Computing.
- Developed a multi-objective fitness function to optimize energy consumption, load balancing, delay, and user allocation rate.
- Introduced a novel hashing-based decoding technique to convert quantum-generated binary solutions into valid AU-ES mappings.
- Conducted experiments using datasets of 50 edge servers and 5000 users, demonstrating improved performance over DE, PSO, and traditional CSO.

TECHNICAL SKILLS AND INTERESTS

Languages:C++, C, HTML, CSS, JavaScript,Python

Developer Tools:VS Code, DEV C++, Git, GitHub, Chrome DevTools

Frameworks: ReactJS, NodeJS

Cloud/Databases: MySQL,MongoDB

Soft Skills: Hardworking, Good Communication Skill,book reading

Coursework:: Data Structure and Algorithms, Operating System,DBMS

Areas of Interest: Software Development, fullstack web development.

POSITIONS OF RESPONSIBILITY

•Position, NSS Volunteer	Position tenure
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ACHIEVEMENTS

- JEE Mains RankHolder** Obtained an All India Rank of 25278 among 15+ Lacs of Candidates.
- Candidate - TCS iON Assessment** Successfully cleared the TCS iON assessment with a High Performance category